

- Create the project directory.
- Initialize the Node.js application.
- Configure npm.
- Install required dependencies.
- Establish the basic backend structure.

Stage 2 – Backend Configuration

- Configure Express.js.
- Create the server.
- Configure middleware.
- Create API routes.
- Configure error handling.

Stage 3 – Database Implementation

- Configure MongoDB.
- Connect MongoDB with Mongoose.
- Create schemas.
- Define models.
- Implement database operations.

Stage 4 – Authentication and Authorization

- Implement user registration.
- Implement login functionality.
- Hash passwords securely.
- Generate JWT tokens.
- Implement role-based access control.

Stage 5 – Blog Management

- Develop blog creation.
- Implement blog retrieval.
 - Implement blog update.
- Implement blog deletion. □
Add categories and tags.

Stage 6 – AI Integration

- Connect the backend with Google Gemini AI.
- Implement AI blog generation.
- Implement AI summarization.
- Implement additional AI-assisted features.

Stage 7 – Testing

- Test API endpoints.
- Test authentication.
- Test authorization.
- Test database operations.
- Test AI functionality.
- Test error handling.

Stage 8 – Documentation and Demonstration

- Prepare project documentation.
- Organize the GitHub repository.
- Capture required project outputs. □ Prepare the demonstration video. □ Submit the final project.

4.3 Project Milestones

Milestone	Activity
M1	Project idea and requirement analysis
M2	System architecture and design
M3	Backend environment setup
M4	Database configuration
M5	Authentication implementation
M6	Blog management implementation
M7	AI feature integration
M8	Testing
M9	Documentation
M10	Project demonstration

4.4 Risk Management

Possible risks include:

- Database connection failures.
- Invalid API requests.
- Authentication errors.
- Unauthorized access.
- AI service failures.
- Incorrect data validation.
- Server configuration problems.

These risks can be reduced through validation, error handling, testing, secure authentication, and proper configuration management.